The empirical recurrence for OEIS A202080, proved from an exact model

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Abstract

OEIS A202080 carries an empirical recurrence of order 17. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an Ehrhart quasi-polynomial, from which a provably satisfies a MONIC linear recurrence of order $S = 19$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A202080 is "Number of arrays of 9 integers in -n..n with sum zero and the sum of every adjacent pair being odd.". It has offset 1 and begins

$$6, 726, 8412, 80812, 340660, 1516410, 4213042, 12861252, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = a(n-1) +8*a(n-2) -8*a(n-3) -28*a(n-4) +28*a(n-5) +56*a(n-6) -56*a(n-7) -70*a(n-8) +70*a(n-9) +56*a(n-10) -56*a(n-11) -28*a(n-12) +28*a(n-13) +8*a(n-14) -8*a(n-15) -a(n-16) +a(n-17)

As of the "Last modified" line on the live entry (Oct 03 2025 09:26:37, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a quasi-polynomial

The array has a FIXED number L of entries and an alphabet that grows with n. Every condition the entry imposes is a boolean combination of statements $\sum_i c_i x_i + c\,n \bowtie 0$ with integer coefficients and $\bowtie$ one of $=, \neq, <, \leq, >, \geq$: the bounds $|x_i| \leq n$ or $0 \leq x_i \leq n$, a vanishing sum, a difference that must not vanish, a pair that must total exactly n. Each is homogeneous in (x, n) jointly, so the admissible x for a given n are precisely the integer points at height n of a finite union of relatively open rational cones in $\mathbb{R}^{L+1}$, and $a(n)$ is the Ehrhart quasi-polynomial of that union: of degree at most L, with period dividing the heights of the cones' ray generators.

3 The bound

The period is derived, not assumed. Every ray of every cell of the arrangement is cut out by L linearly independent hyperplanes taken from the constraint set; writing that $L \times (L+1)$ system as M, its null direction has t-component $\pm\det$ of the x-parts, so the primitive generator (x^*, t^*) has t^* dividing that determinant. Let T be the set of those determinants. A simplicial cone in a triangulation has at most $L+1$ rays, each contributing a factor $1 - z^{t_i}$ with $t_i \in T$, so

$$A(z) \;=\; \prod_{d \,|\, t \text{ for some } t \in T} \Phi_d(z)^{L+1}$$

annihilates a, and $S = \deg A = 19$. A congruence condition modulo m is counted one residue class at a time; a class is a coset of $m\mathbb{Z}^L$, and a generator primitive in $\mathbb{Z}^{L+1}$ must be scaled by a divisor of m to lie in it, so each t is replaced by mt and nothing else changes.

Over the common denominator the numerator has degree below that of A for every cell of the arrangement that HAS a ray, because such a cell's numerator collects a fundamental parallelepiped whose heights are below the sum of its generators' heights. The origin is a cell too and it has none: its series is the constant 1, which over A is A/A and reaches degree exactly $\deg A$. The bound in force is therefore $zA(z)$, of order $S = 19$, and $a(0), \ldots, a(S-1)$, computed exactly, determine every later term. Nothing is fitted, and the annihilator is tested on terms the model was not asked for before anything is claimed — a bound one too small fails that test, which is how this one was found.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^{17} - \sum_i c_i z^{17-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. ☐

Theorem 2.

$$a(n) \;=\; a(n-1)+8\,a(n-2)-8\,a(n-3)-28\,a(n-4)+28\,a(n-5)+56\,a(n-6)-56\,a(n-7)-70\,a(n-8)+70\,a(n-9)+5$$

for every $n > 0$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 0 + 1$ onwards, and the run exceeds $S + 17$, which by Lemma 1 settles every larger index at once. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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